

Synthetic CT Image Generation

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Introduction

We propose a novel framework for generating synthetic chest CT images from free-form anatomical descriptions. The results of this work will be used to augment the training datasets of internal algorithms. Hence, we focus on improving three areas: adherence to **anatomy**, realism of **CT textures**, and coverage of **rare cases**. We experiment with state-of-the-art diffusion architectures and conditioning techniques, evaluating them on a set of fixed benchmarks to validate our workflow.

Method

Our workflow is centered around a primary image diffusion model with conditioning inputs designed around our criteria. We choose NVIDIA's MAISI architecture for this role due to its strong conditioning and pretrained weight availability¹.

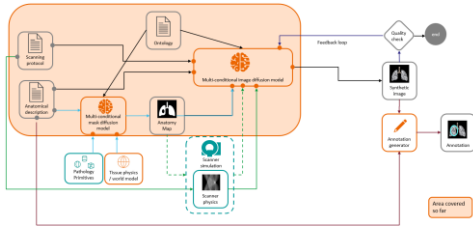


Figure 1: A high-level overview of our envisioned CT image generation workflow, centered around a pre-trained diffusion model with multiple conditioning inputs. The orange box highlights what has been implemented so far.

A major conditioning component is the anatomy map, which when fed into ControlNet² enforces strong adherence to the provided anatomy. MAISI conveniently provides a separate diffusion model that produces input masks for the ControlNet³, so we adapt this to generate stronger variations of anatomy.

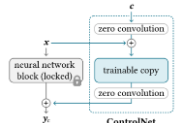


Figure 2: A visualization of how the ControlNet enforces spatial conditioning by building a trainable copy of a diffusion model's encoder blocks.²

To condition both diffusion models, we build a 93-field schema based on a comprehensive chest CT ontology. An LLM parses the free-form report text and metadata and fills in all applicable values. Those values are then embedded based on data type and split into two categories for conditioning injection:

- **Global study properties** (demographics, scanner settings)
 - Added to time embedding
- **Organ-level** findings (organ volume, tumor characteristics)
 - Fed into cross-attention

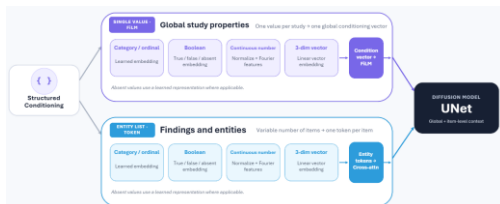


Figure 3: A diagram illustrating how ontology fields are embedded and injected into the diffusion UNet based on data type and category.

Data

Both models are trained over 300 epochs using 9021 standard-dose, angiography, and low-dose thoracic CT images with and without contrast from the Gradient Health and Princeton Radiology datasets. Radiology reports and patient metadata are also included in the datasets, while ground-truth masks are produced by TotalSegmentator.³

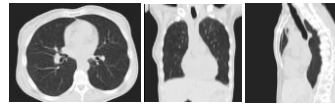


Figure 4: Sample chest CT images from the Gradient Health dataset.



Figure 5: Ground-truth segmentation masks generated by TotalSegmentator from the CT image in Figure 4.

Mask Generation

MAISI's original mask diffusion model is conditioned on a limited schema, with only 10 organ volume controls¹. To help it learn our 93-field ontology, we primarily fine-tune the cross-attention module.

LoRA adapters⁴ (r=16) lightly adjust the frozen query and output projection layers, while key and value layers are trained from scratch for a higher-dimensional schema.

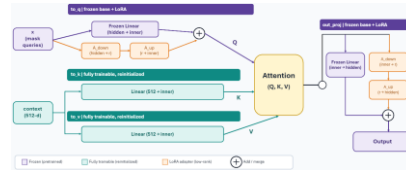


Figure 6: A diagram of the LoRA adapter architecture used to fine-tune the cross-attention module of MAISI's mask diffusion model.

To further enforce our goals of anatomical accuracy and extensibility to rare cases, we add several auxiliary terms to the loss function:

OBJECTIVE	PROMPT	PREDICTION TARGET	TRAINING SIGNAL
1 Morphology	Latent + pretrained model + morphology	Organ volume and bounding boxes	Organ shape descriptors
2 Station	Latent + pretrained model + 4-6 keywords	10-axis occupancy heatmap	Relative spatial descriptors
3 Decoder	Latent patch + VAE decoder + voxel patch	Voxel reconstruction	Ground truth voxel comparison (Exponential loss over 100 steps)
4 Counterfactual	Swap one organ's morphology condition	Counterfactual appearance	Reassess organ color change (Pearson at other changes)

Figure 7: A table of the 4 types of auxiliary loss we add to base mask diffusion loss to refine anatomical accuracy and schema responsiveness.

Image Generation

Since anatomy is strongly controlled by the mask diffusion model and ControlNet, we let the image diffusion model focus on improving CT textures and appearance. Only this relevant subset of fields from the 93-field schema are fed into the model.

The image diffusion model provided by MAISI does not have a pretrained cross-attention module because it didn't take in structured controls. Therefore, we train this module from scratch, following the same structure as Figure 6 but without LoRA adapters.

We also add specialized auxiliary terms to the model, following a similar pattern as the mask diffusion terms:

OBJECTIVE	PROMPT	PREDICTION TARGET	TRAINING SIGNAL
1 Appearance	Latent + pretrained model + FID statistics	Median, Q25, low-attention fraction, etc.	Regional CT intensity/texture
2 Decoder	Latent patch + VAE decoder + voxel patch	Voxel reconstruction	Ground truth voxel comparison (Exponential loss over 100 steps)
3 Counterfactual	Swap one organ's appearance condition	Counterfactual appearance	Reassess organ color change (Pearson at other changes)

Figure 8: A table of the 3 types of auxiliary loss we add to base image diffusion loss to refine CT image appearance and schema responsiveness.

Mask Results

Our evaluation procedure for the mask generation architecture chooses 16 held-out validation cases, 8 from Gradient Health and 8 from Princeton. We then compare two types of metrics: geometrical accuracy and ontology field faithfulness.

We use Dice⁵, an overlap measurement widely used to evaluate segmentation quality, to compute foreground (any organ vs background) and per-organ quality. Below is the formula used to compute Dice (note that scores range between 0 and 1):

$$\text{Dice}(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

where A and B are the voxel sets of the generated and reference masks, respectively.

Table 1: Dice comparison on the 16-case evaluation panel. Delta is the absolute improvement of the fine-tuned mask diffusion model over the baseline.

Metric	Baseline	Fine-Tuned	Δ
Foreground (any-organ)	0.6261	0.8233	+0.1972
left lung upper lobe	0.2188	0.6340	+0.4152
left lung lower lobe	0.1923	0.5787	+0.3864
right lung upper lobe	0.0634	0.5043	+0.5309
right lung middle lobe	0.0082	0.4436	+0.3854
right lung lower lobe	0.0966	0.3928	+0.2962
trachea	0.0204	0.2613	+0.2409
esophagus	0.0215	0.1968	+0.1753
aorta	0.0921	0.4223	+0.3302
heart	0.1195	0.6440	+0.5245
inferior vena cava	0.0633	0.1950	+0.1317
liver	0.0936	0.4199	+0.3263

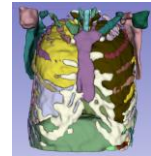


Figure 9: Segmentation masks generated by our fine-tuned mask generator, evaluated against the Figure 5 reference (organs are colored differently due to visualization software)

To examine how faithfully MAISI fine-tuning techniques respond to conditioning, we independently sweep 3 organ-specific fields across 10 values each, then compute a correlation.

Table 2: Faithfulness results for the fine-tuned mask diffusion model. Correlation measures the association between the intervention value and the generated target-label fraction across the 10-point sweep.

Field	Values	Corr.	Injection Method
Body envelope volume:	0.08-0.35	+0.700	Time embedding
Aorta diameter	20-56 mm	+0.027	Cross-attention
Lung tumor volume	0.0005-0.0050	+0.764	Cross-attention

Image Results

To evaluate the final product of our workflow, we feed the generated mask as input to the ControlNet and run inference on our fine-tuned image diffusion model.

On the same 16 cases as above, we compute 2.5D Fréchet Inception Distance (FID)⁶, which extracts features from the reference and generated images using a